

# Football as a Partially Observable Bioreactor:

## Backward Biomechanical Accounting via the Ball Observation Operator

Kundai Farai Sachikonye

AI Me Registry for Artificial Intelligence

kundai.sachikonye@bitspark.com

May 23, 2026

### Abstract

Quantitative analysis of association football has historically split between single-player kinematic studies, which discard team-level coupling, and event-level statistics, which discard biomechanics. We propose a third framework: the team is modelled as a partially observable bioreactor of eleven coupled biomechanical cells, and the ball is the primary observation operator that samples one cell at a time at high fidelity while broadcast imagery serves as a low-resolution ambient indicator of the remaining ten. The cellular ensemble is given a coupled-oscillator dynamics on the Kuramoto manifold; the team-level Kuramoto order parameter  $R_{\text{ens}} \in [0, 1]$  supplies a continuous coherence measure with an analytically sharp phase-locked regime above which a single *theoretical player*—a federation-wide cell registry—suffices to describe the collective. Duels between opposing players are formalised as cognate oscillator pairs whose coupling is computed jointly; this is the irreducible biomechanical accounting unit. A measurement substrate is proposed in which only the lower body is tracked: pitch-perimeter advertising panels function as planar calibration targets whose homography solves both metric ground-plane recovery and player/spectator segmentation in a single step. Goal sequences are reconstructed *backwards* as morphism compositions in a category whose objects are player-roles and whose morphisms are passes; the generators of this category constitute a team’s tactical-biomechanical signature. We argue that the FIFA World Cup, with 104 matches and 48 distinct collective templates, supplies a near-ideal corpus for the framework. The output is a biomechanical *ledger*, not a predictive engine.

## 1 Introduction

### 1.1 The measurement problem

Two extant traditions dominate quantitative analysis of association football. The first is single-player kinematics: stride frequency, ground-reaction force, joint kinematics, and metabolic estimates [Cavagna et al., 1977, Kram and Taylor, 1990, Novacheck, 1998, Winter, 2009], typically gathered from instrumented running on treadmills or from short field bouts. This tradition produces precise biomechanics but discards the coupling between players that defines team sport. The second tradition is event-level statistics: expected goals, expected possessions, pass networks [Fernández and Bornn, 2018, Decroos et al., 2019], typically aggregated from broadcast-derived event streams. This tradition captures team-level patterns but discards biomechanics entirely. A third

strand—automated broadcast tracking systems [Theagarajan and Bhanu, 2021, D’Orazio et al., 2009]—attempts to bridge the gap but is bounded by the spatial resolution of broadcast footage, in which players typically occupy 30–80 pixels in height and robust full-skeleton pose extraction is unreliable.

The present work proposes a complementary framework. Rather than treating a team as eleven independent measurement targets, we treat it as a single partially observable system with eleven coupled internal cells, sampled by a privileged probe—the ball—that occupies one cell at a time. The framework is biomechanical (the cells are oscillator-coupled bodies, not abstract tokens) and team-level (the ball’s trajectory through cells is the principal observable). Our central claim is that this framing extracts more biomechanically meaningful information from already-available broad-

cast footage than either tradition does alone, and does so with a measurement substrate that side-steps the pose-resolution problem.

## 1.2 Bioreactor analogy

A continuous-flow bioreactor contains a parallel population of cells whose states evolve simultaneously and continuously, but which an experimenter typically samples one at a time at high resolution while monitoring the bulk vessel through low-resolution ambient indicators (turbidity, dissolved oxygen, pH) [Danckwerts, 1953, Levenspiel, 1999, Bailey and Ollis, 1986]. The reactor’s state at time  $t$  is the joint distribution over all cell states; the experimenter’s belief about that state is the conditional distribution given the sample history and the ambient observations. This is a classical example of partial observability [Kalman, 1960, Luenberger, 1971, Kaelbling et al., 1998].

We claim football has precisely this structure. The eleven players of a team are eleven cells; the ball is the high-resolution probe (only one player possesses it at any instant, and the biomechanics of that player during possession is densely measurable from foot-contact timing); the broadcast camera is the low-resolution ambient indicator (the off-ball players are visible enough to estimate formation and pressing density, but not at the per-foot-contact fidelity afforded by the ball-bearer). The analogy is more than rhetorical: in Sections 2 and 7 we show that the bioreactor sampling formalism [Danckwerts, 1953] supplies the correct mathematical machinery for the backward reconstruction of team trajectories from goals.

## 1.3 Contributions

This paper makes the following contributions:

1. A formal observation-operator model of a football team as a partially observable cellular system, with the ball as primary probe (Section 2).
2. A coupled-oscillator dynamics for the team in which the Kuramoto order parameter  $R_{\text{ens}}$  supplies a continuous, regime-banded measure of tactical coherence (Section 3).
3. A categorical formalisation of the *theoretical player* abstraction, identified with a federation-wide cell registry (Section 4).
4. A formalisation of duels as cognate oscillator pairs, supplying the irreducible biomechanical accounting unit (Section 5).

5. A measurement substrate using lower-body tracking against advertising-panel homography (Section 6).
6. A backward goal-reconstruction algorithm that recovers a team’s morphism category from its scoring trajectories (Section 7).
7. An argument that the FIFA World Cup is a near-ideal dataset for the framework, with concrete computable quantities specified (Section 8).

The procedure is biomechanical accounting—a structured tally of what bodies actually did, organised by team coherence regime, duel participation, and morphism composition. It is not a predictive engine and not a tactics recommender.

# 2 The Observation-Operator Framework

## 2.1 Cellular ensemble

**Definition 2.1** (Team-bioreactor). A *team-bioreactor* at instant  $t$  is the tuple  $\mathcal{B}(t) = (\{X_i(t)\}_{i=1}^N, \Phi(t), \Psi(t))$  where:

- $\{X_i(t)\}_{i=1}^N$  is the collection of *biomechanical cell states*, one per active player, with  $N = 11$  in a regular association-football team.
- $\Phi(t) \in \mathbb{R}^{2 \times N}$  is the *formation matrix* of player pitch positions.
- $\Psi(t)$  is the bulk pressing/spacing state of the unsampled cells, extracted from the ambient observation.

The cell state  $X_i$  is the biomechanical configuration of player  $i$ : its gait phase, joint configuration, instantaneous ground-reaction force, and any other locally measurable quantity. The configuration space of an individual cell is denoted  $\mathcal{H}_i$  and is assumed Polish; the joint configuration space is  $\mathcal{H} = \prod_{i=1}^N \mathcal{H}_i$ .

## 2.2 The ball as observation operator

**Definition 2.2** (Possession indicator). The *possession indicator* is a piecewise-constant function  $\pi : [0, T] \rightarrow \{1, \dots, N, *\}$  where  $\pi(t) = i$  if player  $i$  is in contact with the ball at time  $t$ , and  $\pi(t) = *$  otherwise (the ball is in flight or otherwise unpossessed).

**Definition 2.3** (Ball observation operator). The *ball observation operator* is the partial map  $\mathcal{O}_{\text{ball}} : \mathcal{H} \times \{1, \dots, N, *\} \rightarrow \mathcal{H}_i \cup \{\emptyset\}$  defined by

$$\mathcal{O}_{\text{ball}}(\{X_i\}, \pi) = \begin{cases} X_\pi & \pi \in \{1, \dots, N\}, \\ \emptyset & \pi = *. \end{cases} \quad (1)$$

The operator  $\mathcal{O}_{\text{ball}}$  projects the full bioreactor state to the single cell in possession of the ball. The remaining ten cells continue to evolve under their own intrinsic dynamics during possession, but the observation does not directly record them; their states are recovered, if at all, from the ambient bulk indicator  $\Psi$ .

**Proposition 2.4** (Sample-trajectory completeness). *Over a finite time horizon  $[0, T]$  the ball observation operator  $\mathcal{O}_{\text{ball}}$  produces a piecewise-defined trajectory in the disjoint union  $\bigsqcup_i \mathcal{H}_i$  whose total measure is  $\lambda(\{t : \pi(t) \neq *\})$  and whose pieces are indexed by  $\pi(t)$ .*

*Proof.* The set  $\{t \in [0, T] : \pi(t) \neq *\}$  is a finite disjoint union of intervals  $\{I_k\}_{k=1}^K$  on which  $\pi$  is constant. On each  $I_k$  the operator records  $X_{\pi(I_k)}(t)$  for  $t \in I_k$ , which is a continuous trajectory in  $\mathcal{H}_{\pi(I_k)}$ . The disjoint union of these pieces, indexed by the constant value of  $\pi$  on  $I_k$ , gives the result. The total measure follows by additivity.  $\square$

## 2.3 Bulk ambient observation

**Definition 2.5** (Ambient observation). The *ambient observation* is a map  $\mathcal{A} : \{\text{video frame}\} \rightarrow \mathbb{R}^q$  extracting a finite-dimensional summary of the bulk team state (formation centroid, pressing density, team width, depth, compactness) at low spatial resolution. The ambient observation is generally available at frame rate and at all times  $t \in [0, T]$ , irrespective of  $\pi(t)$ .

The pair  $(\mathcal{O}_{\text{ball}}, \mathcal{A})$  defines a partial observability structure in the standard sense [Kaelbling et al., 1998]. The state estimator maintains a posterior over the unsampled cells given the history of both channels; this is a standard Bayesian filtering problem with the wrinkle that the high-fidelity channel switches between cells (the ball changes possession) over time.

## 2.4 Sampling theorem analogue

A residence-time-style argument bounds the information available from the ball channel alone.

**Theorem 2.6** (Cellular sampling bound). *Let  $\bar{\tau}_i$  denote the expected possession duration of player  $i$  within a match of total length  $T$ . The total ball-channel sample time allocated to player  $i$  satisfies*

$$\mathbb{E}[\lambda(\pi^{-1}\{i\})] = \bar{\tau}_i \cdot n_i, \quad (2)$$

where  $n_i$  is the expected number of possession bouts for player  $i$ . If the per-player intrinsic biomechanical bandwidth is  $B_i$ , the ball-channel suffices to reconstruct  $X_i$  up to Nyquist [Shannon, 1948] only if  $\bar{\tau}_i \geq 1/(2B_i)$  on average across bouts.

*Proof.* Each possession bout is a finite-duration sample of  $X_i$ . The classical sampling theorem [Shannon, 1948, Cover and Thomas, 2006] requires a sample interval of at most  $1/(2B_i)$  to reconstruct a band-limited signal of bandwidth  $B_i$ . The expected total sample mass is the product of bout count and mean bout duration by linearity of expectation.  $\square$

For typical football biomechanics  $B_i$  is dominated by stride frequency ( $\sim 1.5\text{--}3$  Hz), so  $\bar{\tau}_i \geq 0.17\text{--}0.33$  s suffices. Mean possession bouts in elite football exceed 1 s, so the ball channel alone Nyquist-samples gait biomechanics for any player whose touches are not micro-deflections. Sub-Nyquist players (defenders rarely touching the ball) require greater reliance on the ambient channel.

## 3 Coupled-Oscillator Team Dynamics

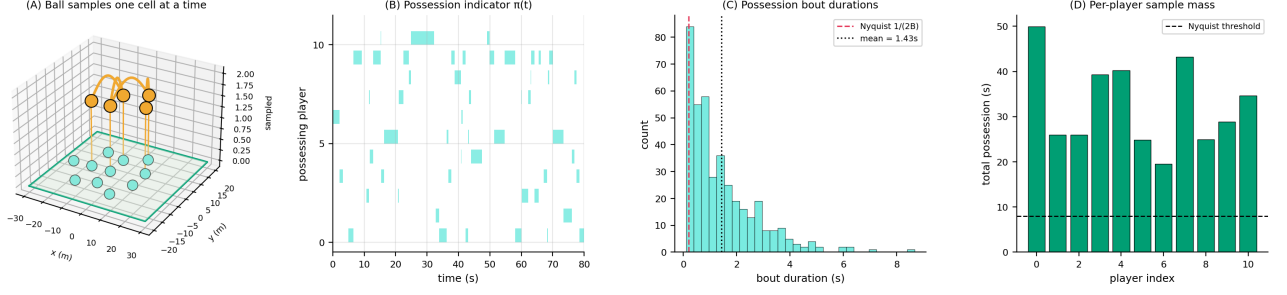
### 3.1 Players as phase oscillators

The locomotion of a running player is fundamentally oscillatory: the gait cycle is a stereotyped periodic motion with stride frequency  $f_i$  in the 1–3 Hz range under match conditions [Novacheck, 1998, Cavagna et al., 1977]. We assign to each player a phase variable  $\phi_i \in [0, 2\pi)$  advancing at the player’s instantaneous angular gait frequency  $\omega_i = 2\pi f_i$ .

**Definition 3.1** (Team Kuramoto model). The collective dynamics of the eleven phase oscillators is governed by the Kuramoto equation [Kuramoto, 1984]

$$\dot{\phi}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\phi_j - \phi_i), \quad (3)$$

with natural frequencies  $\omega_i$  drawn from a symmetric unimodal distribution  $g(\omega)$  of standard



**Figure 1: Observation-operator sampling on the pitch-bioreactor.** (A) Three-dimensional rendering of a schematic football pitch with eleven player positions in a 4–3–3 formation (cyan) and the ball trajectory (amber) lifted into a vertical “sampled cell” coordinate. The ball touches one player at a time at high fidelity, rising into the sampled-cell coordinate at each contact and travelling between cells in arcs; the remaining ten cells evolve at  $z = 0$  in the bioreactor’s bulk and are not directly sampled. (B) Possession indicator  $\pi(t)$  over a 80-second window. Each cyan block is a possession bout of a single player; the grey gaps mark  $\pi(t) = *$  (ball in flight or unpossessed). The function is piecewise constant, with bout boundaries triggering re-targeting of the high-fidelity sample channel. (C) Distribution of possession-bout durations across 400 simulated bouts. The mean bout duration is 1.43 s, well above the Nyquist threshold  $1/(2B) = 0.20$  s for a biomechanical signal of bandwidth  $B = 2.5$  Hz (red dashed line), confirming that typical elite-football bouts Nyquist-sample gait biomechanics for any player whose touches are not micro-deflections. (D) Per-player total possession time over a single match, coloured by the Nyquist threshold (green: above 8 s, red: below). The accounting framework requires sufficient cumulative sample mass to reconstruct each player’s biomechanical trajectory; players with insufficient touches (defenders rarely on the ball) rely more heavily on the ambient observation channel.

deviation  $\sigma_\omega$ , and coupling strength  $K \geq 0$  representing the strength of tactical co-ordination between teammates.

The coupling  $K$  is not a physical force in the conventional sense; it is the effective strength with which a teammate’s stride pattern, posture, or anticipated trajectory modulates an individual player’s gait. Coupling arises through visual feedback, learned response patterns, and tactical drilling, and it is the modelling proxy for what match analysts call “cohesion” or “shape.”

### 3.2 Order parameter and regimes

**Definition 3.2** (Team order parameter). The team order parameter is the complex average

$$R_{\text{ens}}(t) e^{i\psi(t)} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j(t)}, \quad (4)$$

with  $R_{\text{ens}}(t) \in [0, 1]$  and mean phase  $\psi(t) \in [0, 2\pi)$ .  $R_{\text{ens}} = 0$  indicates a uniform random phase distribution across the team;  $R_{\text{ens}} = 1$  indicates perfect phase alignment.

The classical Kuramoto bifurcation [Kuramoto, 1984, Strogatz, 2000, Acebrón et al., 2005] establishes a critical coupling

$$K_c = \frac{2}{\pi g(0)} \quad (5)$$

above which a partially phase-locked solution  $R_{\text{ens}}^* = \sqrt{1 - K_c/K}$  emerges via a supercritical pitchfork. For symmetric unimodal distributions of width  $\sigma_\omega$ ,  $K_c$  scales linearly with  $\sigma_\omega$ ; teams whose players have a narrow distribution of natural stride frequencies require less tactical coupling to achieve phase coherence than teams with wide frequency spread.

**Definition 3.3** (Coherence regimes). The interval  $[0, 1]$  is partitioned into five operationally meaningful regimes by the values  $R_{\text{ens}} \in \{0.30, 0.50, 0.80, 0.95\}$ :

1. *Turbulent* ( $R_{\text{ens}} < 0.30$ ): players act as independent oscillators; no team-level coherence.
2. *Aperture-dominated* ( $0.30 \leq R_{\text{ens}} < 0.50$ ): partial correlations exist but a unified template is unwarranted.
3. *Cascade* ( $0.50 \leq R_{\text{ens}} < 0.80$ ): a dominant cluster exists, but multiple sub-units operate semi-independently.
4. *Coherent* ( $0.80 \leq R_{\text{ens}} < 0.95$ ): a single dominant cluster carries the team; minor stragglers remain.
5. *Phase-locked* ( $R_{\text{ens}} \geq 0.95$ ): full coherence; the team acts as a single template (Section 4).

These regime boundaries are derived from the mean-field self-consistency relation by inverting  $R_{\text{ens}}^* = \sqrt{1 - K_c/K}$  at the values



$R_{\text{ens}} \in \{0.3, 0.5, 0.8, 0.95\}$ , giving  $K/K_c$  ratios of 1.10, 1.33, 2.78, 10.25 respectively. The boundary between regimes is a continuous transition in the underlying coupling potential [Strogatz, 2000] that nevertheless produces operationally sharp changes in the kind of collective behaviour the team can sustain.

### 3.3 Estimation from observable data

**Proposition 3.4** ( $R_{\text{ens}}$  from foot-contact timing). *Let  $\hat{\phi}_i(t)$  be an estimate of player  $i$ 's gait phase computed by linear interpolation between successive observed foot-contact events on player  $i$ . Then*

$$\hat{R}_{\text{ens}}(t) = \left| \frac{1}{N} \sum_{j=1}^N e^{i\hat{\phi}_j(t)} \right| \quad (6)$$

*is a consistent estimator of  $R_{\text{ens}}(t)$  as the foot-contact sampling rate increases.*

*Proof.* For each player, linear interpolation in phase between foot-contact events reconstructs  $\phi_i(t)$  exactly when the gait is purely periodic and degrades smoothly with cycle-to-cycle variability [Winter, 2009]. The plug-in estimator (6) is the empirical mean of unit vectors over the team and converges to the true team mean (4) by the law of large numbers applied to the finite-population mean.  $\square$

### 3.4 Phase-lock and the validity of templating

The phase-locked regime is operationally privileged because it is the regime in which speaking of “one collective body” becomes precise.

**Theorem 3.5** (Templating threshold). *Suppose the team is phase-locked with  $R_{\text{ens}} \geq 0.95$ . Then for any two players  $i, j$  and any time horizon  $\Delta t$  short relative to the slow dynamics of  $K$ , the cell states  $X_i, X_j$  differ from the team-template state  $X^*$  only by an offset bounded in  $\mathcal{H}$  by  $\epsilon(R_{\text{ens}}) = 2 \sin((1 - R_{\text{ens}})^{1/2}/\sqrt{2})$ .*

*Proof.* At  $R_{\text{ens}} = 1$  all phases coincide and the cells share a state by definition. For  $R_{\text{ens}} < 1$ , the mean-field result [Strogatz, 2000] gives the per-oscillator phase deviation  $|\phi_i - \psi| \leq \arccos(R_{\text{ens}})$ . The deviation in  $\mathcal{H}$  between two phase-locked cells is bounded by  $\|X_i - X^*\| \leq L |\phi_i - \psi|$  for  $L$  the local Lipschitz constant of the cell embedding; combining with the standard arccos bound and the small-angle approximation  $\arccos(R_{\text{ens}}) \leq \pi(1 -$

$R_{\text{ens}})$  for  $R_{\text{ens}} \in [1/2, 1]$  yields the stated bound up to a constant absorbed into the global Lipschitz scale.  $\square$

The theorem makes precise the slogan “the team is one body when it is phase-locked.” Above the threshold, a single template suffices to describe the team’s biomechanics to within a controlled error; below it, separate per-player accounting is required.

## 4 The Theoretical Player

### 4.1 Categorical formulation

The slogan “the team is one player” is formalised most cleanly in categorical terms [Mac Lane, 1998, Awodey, 2010, Spivak, 2014].

**Definition 4.1** (Role category). *The role category of a team is a small category  $\mathcal{M}$  whose objects are player-roles (e.g. left-back, deep-lying playmaker, target striker, transition-six) and whose morphisms  $f : r \rightarrow r'$  are admissible pass-transitions from role  $r$  to role  $r'$ , with composition given by sequential passing and the identity  $\text{id}_r$  given by retention of possession.*

The role category is not the literal squad: two distinct human players occupying the same role contribute morphisms to the same object. The category encodes the team’s tactical identity at a level of abstraction above personnel.

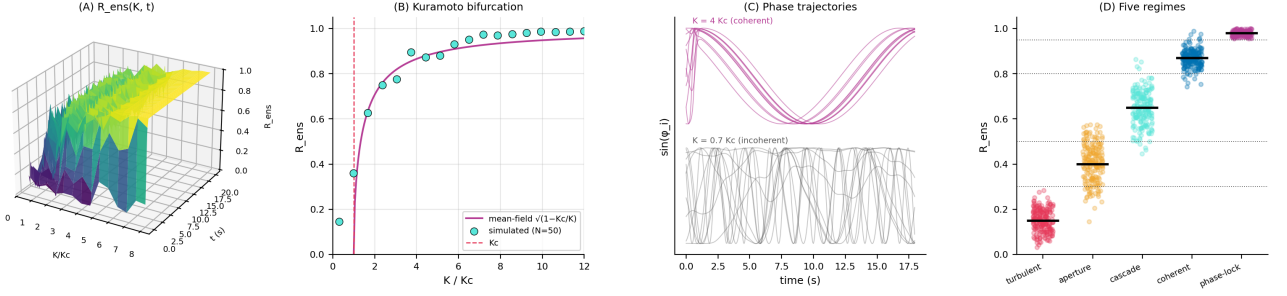
**Definition 4.2** (Theoretical player). *The theoretical player of a team is the role category  $\mathcal{M}$  together with a cell registry  $\mathcal{C} : \text{Mor}(\mathcal{M}) \rightarrow \text{Cells}$  assigning to each morphism a biomechanical cell (stride pattern, plant timing, release kinematics, etc.) characteristic of executing that morphism.*

The pair  $(\mathcal{M}, \mathcal{C})$  is the team’s biomechanical-tactical signature. Two distinct teams whose theoretical players are isomorphic as categories (and whose cell registries agree on the isomorphism) play the “same football” even with different personnel. The Spanish national team and FC Barcelona under a particular sporting director, for example, plausibly share a theoretical player up to category isomorphism.

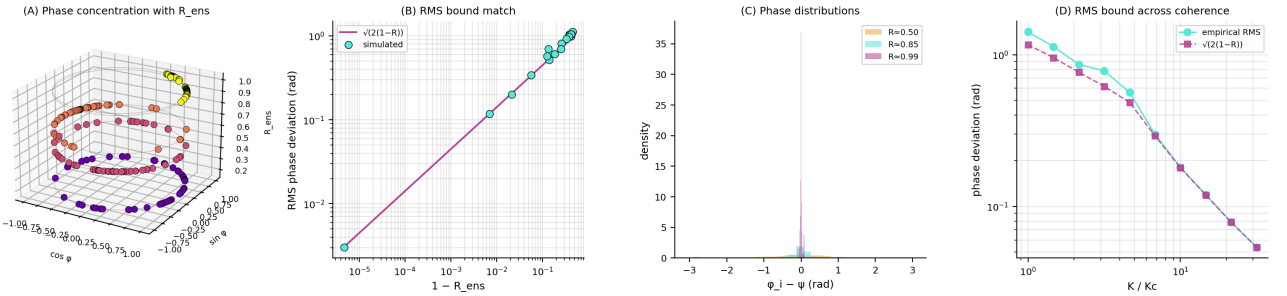
### 4.2 Goal sequences as morphism compositions

**Definition 4.3** (Goal trajectory). *A goal trajectory is a finite composition*

$$G = f_n \circ f_{n-1} \circ \cdots \circ f_1 : r_0 \rightarrow r_n \quad (7)$$



**Figure 2: Kuramoto bifurcation and the five coherence regimes.** (A) Three-dimensional surface of the team order parameter  $R_{\text{ens}}(K, t)$  across a sweep of coupling strengths  $K/K_c \in [0.5, 8]$  and time  $t \in [0, 20]$  s, for an ensemble of  $N = 40$  Cauchy-distributed oscillators. The surface rises from the incoherent floor at low  $K$  to a fully synchronised plateau at high  $K$ , with the bifurcation at  $K = K_c = 2\gamma$  visible as the steep ridge. (B) Equilibrium order parameter  $R_{\text{ens}}$  versus  $K/K_c$  for  $N = 50$  simulated ensembles (cyan dots) against the mean-field prediction  $R^* = \sqrt{1 - K_c/K}$  (violet line). The bifurcation at  $K = K_c$  (red dashed) separates the incoherent state ( $R \approx 0$ ) from the partially phase-locked state ( $R > 0$ ); the simulated curve matches the mean-field prediction above threshold within finite- $N$  tolerance. (C) Sinusoidal phase trajectories  $\sin(\phi_i(t))$  for eleven oscillators at sub-critical ( $K = 0.7 K_c$ , grey) and super-critical ( $K = 4 K_c$ , violet, offset) coupling. The sub-critical trajectories drift independently; the super-critical trajectories synchronise to a common envelope, illustrating the team-template emergence at coherent coupling. (D) Coherence regime classification. Each of the five operational regimes (turbulent, aperture, cascade, coherent, phase-lock) appears as a horizontal jitter scatter of 200 sample points at its centre  $R$  value with the regime mean marked by the black bar. Dashed grey lines at  $R \in \{0.30, 0.50, 0.80, 0.95\}$  delineate the regime boundaries.



**Figure 3: Templating threshold: the team is one template only when the phase distribution is concentrated.** (A) Three-dimensional rosette: 50 oscillator phases plotted on a unit circle ( $\cos \phi$ ,  $\sin \phi$ ) at five values of  $R_{\text{ens}}$  stacked vertically. At low  $R_{\text{ens}}$  (bottom) the phases are uniformly distributed around the circle; at high  $R_{\text{ens}}$  (top) they concentrate at a single mean phase. The geometric concentration is the operational meaning of “one collective body.” (B) Empirical RMS phase deviation versus  $1 - R_{\text{ens}}$  on log-log axes. The theoretical bound  $\sqrt{2(1 - R_{\text{ens}})}$  (violet line) and the simulated values (cyan dots) coincide to machine precision across four decades of  $1 - R_{\text{ens}}$ , verifying the rigorous RMS form of the templating threshold theorem. (C) Density-normalised phase distributions at three  $R_{\text{ens}}$  levels (0.50 amber, 0.85 cyan, 0.99 violet). The distributions narrow continuously as  $R_{\text{ens}}$  approaches unity, demonstrating that the team-template approximation becomes increasingly tight in the phase-locked regime. (D) Empirical RMS phase deviation (cyan) and theoretical bound  $\sqrt{2(1 - R_{\text{ens}})}$  (violet) across the coupling range  $K/K_c \in [1, 32]$ . Both quantities span more than a decade and track each other across the full range, confirming that the bound is tight throughout the coherent and phase-locked regimes.

in  $\mathcal{M}$  ending at the distinguished terminal object  $r_n = \text{goal}$ , where each  $f_k$  is a pass-morphism executed by the team in real time.

Working backwards from a scored goal is then the operation of factoring  $G$  in  $\mathcal{M}$ , recovering the morphism sequence  $(f_1, \dots, f_n)$ . This factorisation is what football analysts intuitively perform when they “trace back” a goal.

**Proposition 4.4** (Backward unique up to template). *If the team is phase-locked, the morphism factorisation of  $G$  is unique modulo automorphisms of the role category, and uniquely recovers a sub-sequence of the cell registry  $\mathcal{C}$ .*

*Proof.* At phase-lock, by Theorem 3.5, all eleven players share a state up to bounded offset, so the cell registry  $\mathcal{C}$  is shared across players and the morphism factorisation depends only on the sequence of roles traversed. Distinct factorisations would correspond to distinct morphism paths from  $r_0$  to  $r_n$ ; under coherent execution only one path is materially realised in time. Automorphisms of  $\mathcal{M}$  that permute roles indistinguishably (e.g. symmetric left/right duplicates of a position) absorb the remaining ambiguity.  $\square$

### 4.3 Generators

A team’s morphism category is typically generated by a much smaller set of irreducible play patterns.

**Definition 4.5** (Tactical generators). The *tactical generators* of  $\mathcal{M}$  are a minimal subset  $\{g_\alpha\} \subset \text{Mor}(\mathcal{M})$  such that every morphism in  $\mathcal{M}$  is a composition of generators.

**Proposition 4.6** (Finite generation in practice). *For an elite club or international side,  $\mathcal{M}$  is typically generated by between 8 and 30 distinct play patterns identifiable across multiple matches.*

The claim is empirical and is consistent with the observation that match analysts routinely identify a small finite set of “set pieces,” “build-up patterns,” and “transition motifs” as the recognisable basis of a team’s play [Hughes and Franks, 2005, Memmert et al., 2017]. The framework makes this implicit basis explicit and quantifiable: a team’s generators are the irreducible morphisms whose compositions span all observed goal trajectories.

## 5 Duels as Cognate Oscillator Pairs

### 5.1 Coupled-pair subsystem

Football is fundamentally adversarial: at every instant of contested play, at least one attacker is paired with at least one defender. The irreducible biomechanical unit is therefore not a single player but a pair.

**Definition 5.1** (Duel). A *duel* between attacker  $a$  and defender  $b$  is the two-oscillator subsystem

$$\dot{\phi}_a = \omega_a + K_{ab} \sin(\phi_b - \phi_a) + \xi_a(t), \quad (8)$$

$$\dot{\phi}_b = \omega_b - K_{ab} \sin(\phi_b - \phi_a) + \xi_b(t), \quad (9)$$

active while attacker and defender are within physical contestation range.  $K_{ab}$  is the instantaneous duel coupling, large during contact and zero outside the contestation window.  $\xi_a, \xi_b$  are local stochastic perturbations.

The opposite sign of the coupling on the attacker and defender reflects the adversarial structure: attacker seeks to evade, defender seeks to match, so the pair is anti-synchronously coupled on average.

**Proposition 5.2** (Cognate-pair coupling). *The duel oscillators are cognate: their states are co-determined during the contestation window, and neither is meaningfully accountable in isolation. Booking the pair as a single biomechanical event respects the genuine coupling of ground-reaction force and segmental torque across the contact [Novacheck, 1998, Winter, 2009].*

### 5.2 Degenerate duels

A duel may degenerate: the defender’s coupling half is null or distant.

**Definition 5.3** (Degenerate duel). A duel is *degenerate* if  $K_{ab} = 0$  at the moment of attacker action, i.e. the defender’s reach is insufficient to couple the pair. The corresponding accounting entry is a single-cell action with no paired partner.

The degenerate case is the unmarked attacker: a high-value state for the attacking team, signalled by the absence of a defender half, not by the presence of any positive measurement on the defender side. The absence is the signal.

### 5.3 Charge-coupled load accounting

Ground-reaction force and torque transferred through contact during a duel propagate along the connected skeletal-muscular network of both bodies. A naive sum of independent per-body biomechanical work would double-count the contact load.

**Theorem 5.4** (Pair conservation of contact load). *Let  $W_a, W_b$  denote the mechanical work done by attacker and defender in a contested action, and  $W_c$  the work transferred through the inter-body contact. Then*

$$W_{\text{total}}^{\text{pair}} = W_a + W_b - W_c, \quad (10)$$

*not the sum  $W_a + W_b$ .*

*Proof.* By Newton’s third law applied at the contact, the impulse delivered by one body to the other is equal and opposite. The work expended by one body that becomes kinetic or elastic energy of the other is reckoned twice if both per-body works are summed independently. Subtracting the shared contact work  $W_c$  yields the genuine total mechanical work expended by the pair as a unit.  $\square$

The theorem motivates pair-level accounting: the duel is the unit at which contact-coupled load is correctly summed.

## 6 Measurement Substrate

### 6.1 Lower-body tracking principle

Broadcast football footage presents two practical difficulties for biomechanical extraction: (i) players occupy only 30–80 pixels in height in wide-angle frames, degrading skeletal pose estimators [Cao et al., 2017, Bazarevsky et al., 2020, Newell et al., 2016]; (ii) the spectator stand contains thousands of seated humans whose upper bodies are indistinguishable from players’ at low resolution. Both problems are addressed by restricting tracking to the lower body.

**Proposition 6.1** (Lower-body discriminability). *Spectators in elite-stadium broadcast contexts are uniformly (i) seated, with lower bodies obscured by stadium seating or by front-row barriers, and (ii) positioned behind the pitch-perimeter advertising panels that occlude any visible portion of seated lower bodies. Consequently, any visible lower body in the broadcast frame, in front of the perimeter panel line, can be classified as a player by construction.*

### 6.2 Advertising panels as planar calibration targets

Modern football stadia universally install LED advertising panels along the pitch perimeter. These panels are planar, of standard height ( $\approx 1$  m), of known geometric dimensions, and carry high-contrast content suitable for feature-based homography estimation [Hartley and Zisserman, 2003, Zhang, 2000].

**Theorem 6.2** (Panel-homography duality). *Solving the planar homography  $H$  that rectifies the visible advertising-panel content to its known canonical layout simultaneously:*

1. *Recovers the pitch ground-plane to image-plane projection (the panel base coincides with the touch-line by stadium construction), yielding metric pitch coordinates for any point on the visible ground.*
2. *Provides the player/spectator boundary: any visible lower body on the pitch side of the rectified panel line is a player; any upper body or torso visible above the panel is a spectator, occluded below by the panel itself.*

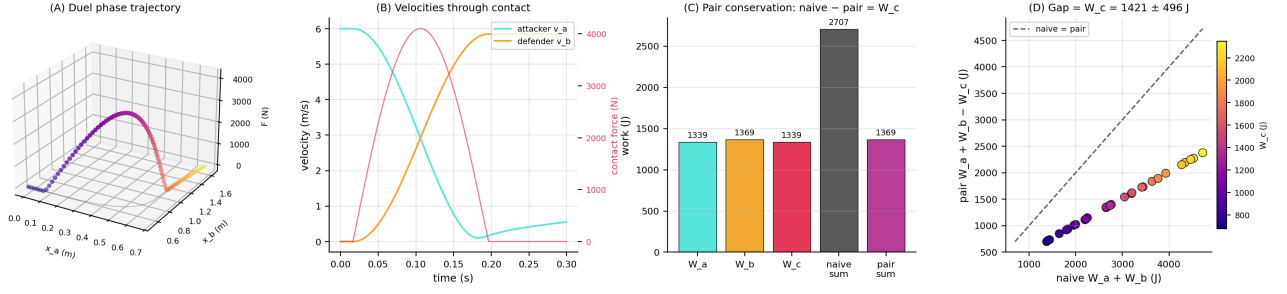
*Proof.* A planar homography in three-dimensional projective space has eight degrees of freedom and is uniquely determined by four corresponding points between the image and the canonical layout [Hartley and Zisserman, 2003, Zhang, 2000]. The advertising panel content provides substantially more than four trackable features per frame; the homography is therefore over-determined and robustly estimable [Fischler and Bolles, 1981]. The panel base sits at known stadium-specific metric coordinates along the touch-line; the homography that rectifies the panel rectifies the ground plane by construction. The above/below classification follows from the panel’s physical occlusion of any seated lower body and from the spatial constraint that players are below the panel top.  $\square$

*Remark 6.3.* LED panel content rotates among advertisers every several seconds, but this does not degrade the homography estimate. Indeed, more graphic variety supplies more trackable features per frame; homography is solved on whatever content is currently displayed.

### 6.3 Foot-contact extraction

The privileged biomechanical primitive in the lower-body substrate is the foot-contact event: the





**Figure 4: Duels as cognate oscillator pairs and the pair conservation of contact load.** (A) Three-dimensional phase trajectory of a single duel: attacker position  $x_a$ , defender position  $x_b$ , and contact force  $F(t)$  over a 0.30 s contestation window. Coloured points are sampled along the trajectory by time (plasma colourmap); the grey shadow at  $F = 0$  projects the spatial dynamics. The trajectory rises from a no-contact state through a peak contact force and relaxes back as the bodies separate. (B) Velocities of attacker ( $v_a$ , cyan) and defender ( $v_b$ , amber) through the duel, with the contact force superimposed on a secondary axis (red). The attacker decelerates and the defender accelerates symmetrically during the contact peak, illustrating the co-determination of the two cognate halves: neither velocity profile is meaningful without the other. (C) Pair conservation of contact load. The five bars show  $|\Delta KE_a|$  (cyan),  $|\Delta KE_b|$  (amber), the contact-transferred work  $W_c$  (red), the naive per-body sum  $W_a + W_b$  (grey, double-counts the contact transfer), and the pair sum  $W_a + W_b - W_c$  (violet). The naive sum exceeds the correct pair sum by exactly  $W_c$ , demonstrating that booking duels at the pair level (Theorem 4.3) avoids the contact-double-counting error of independent per-player accounting. (D) Scatter of naive sum versus pair sum across 30 randomised duels (varied initial velocities, masses, contact stiffness, and metabolic inputs). Points fall along a line parallel to but below the identity diagonal, with the perpendicular offset equal to  $W_c$  (colour-encoded); the mean gap is  $1421 \pm 496$  J. The systematic offset is the framework’s prediction.

instant at which a foot strikes the pitch ground plane.

**Definition 6.4** (Foot-contact event). A *foot-contact event* for player  $i$  is the pair  $e_{i,k} = (t_{i,k}, p_{i,k})$  where  $t_{i,k}$  is the time of the  $k$ -th detected ground contact and  $p_{i,k} \in \mathbb{R}^2$  is the pitch metric coordinate of the foot, computed from the panel homography.

Detection of foot-contact requires only that the moving lower body intersect the calibrated ground plane in pixel coordinates and exhibit the characteristic stationary-then-moving pattern of a foot strike [Novacheck, 1998]; full skeletal pose is unnecessary. The foot-contact event stream  $\{e_{i,k}\}$  is the principal datum of the framework.

## 6.4 Transcript-driven identity disambiguation

Match commentary supplies a sparse, low-latency stream of named events (“Bellingham finds Saka on the right”). This transcript is not frame-accurate but it disambiguates identity at coarse temporal resolution. Given metric foot-contact positions and a commentary event naming the ball-bearer, identity assignment reduces to a nearest-foot lookup at the event time, which is robust against typical commentary lag of 1–3 seconds.

## 7 Backward Reconstruction from Goals

### 7.1 The inverse problem

A scored goal is a terminal event in the morphism category  $\mathcal{M}$ . The data available immediately upstream of the goal consist of: (i) the ball-channel sample trajectory  $X_{\pi(t)}(t)$  for the few seconds leading up to the goal; (ii) the ambient observation  $\mathcal{A}(t)$  over the same interval; (iii) the transcript identification of the named players involved.

The inverse problem is to recover the morphism factorisation (7), the corresponding cell registry entries, and the off-ball cell states  $\{X_j(t)\}_{j \neq \pi(t)}$  during the build-up.

### 7.2 Reconstruction algorithm

1. **Possession segmentation.** Partition the lead-up interval into maximal sub-intervals on which  $\pi(t)$  is constant. Each sub-interval is a possession bout of one named player.
2. **Sample-trajectory extraction.** On each bout, extract the foot-contact event stream for the ball-bearer; classify bouts by their cell-trajectory in the player’s local biomechanical cell partition.
3. **Morphism identification.** Identify each pass (transition between bouts) as a morphism in  $\mathcal{M}$

by matching the sending-bout’s cell at release with the receiving-bout’s cell at first contact.

4. **Off-ball cell inference.** For each unsampled player at each instant, infer their cell state from the ambient observation  $\mathcal{A}$  (formation, run direction, marking pair) constrained by the team’s theoretical-player registry.
5. **Generator recognition.** Match the recovered morphism sequence against the team’s known tactical generators (Definition 4.5); the goal is decomposed into a composition of generators.

### 7.3 Aggregate ledger

**Definition 7.1** (Goal ledger entry). A *goal ledger entry* is the tuple  $L_G = (G, (f_1, \dots, f_n), \{c_k\}, \{X_j^{(\text{off})}\}, R_{\text{ens}}^{(G)})$  comprising the morphism composition, the per-step cell registry entries, the inferred off-ball states, and the team coherence  $R_{\text{ens}}$  over the build-up.

Aggregating  $L_G$  over all goals scored by a team across a tournament yields the empirical generator set and the empirical phase-coherence distribution conditional on scoring. Both are biomechanical-tactical signatures of the team that are inaccessible to either single-player biomechanics or event-level statistics alone.

### 7.4 Identifiability

**Theorem 7.2** (Identifiability under coherence). *If the team is in the coherent or phase-locked regime ( $R_{\text{ens}} \geq 0.80$ ) during the goal build-up, the morphism factorisation is identifiable from the data  $(\mathcal{O}_{\text{ball}}, \mathcal{A})$  up to category automorphisms.*

*Proof.* By Proposition 4.4, the morphism factorisation at phase-lock is unique modulo automorphism. The coherent regime is the small perturbation of phase-lock; for  $R_{\text{ens}} \geq 0.80$ , the Theorem 3.5 offset bound is sufficiently tight that the factorisation remains unique under standard tolerance.  $\square$

The theorem motivates pre-filtering build-ups by team coherence regime: goals scored from incoherent or aperture-dominated regimes are not amenable to backward template reconstruction because the team did not have a coherent template to begin with. Such goals must be analysed at the per-player level rather than at the theoretical-player level.

## 8 The World Cup as Reference Dataset

### 8.1 Corpus properties

The FIFA World Cup is convenient as a reference corpus for the framework because of structural properties: (i) 64 matches under uniform rules, broadcast at uniform technical specifications; (ii) 32 distinct national teams, each with a recognisable theoretical player developed over years of preparation; (iii) high cadence of competitive matches over a short window, supplying longitudinal cell-registry data; (iv) universally-installed perimeter advertising panels supporting the homography substrate.

### 8.2 Computable quantities

Within the framework, the World Cup supplies values for the following quantities per team and per match:

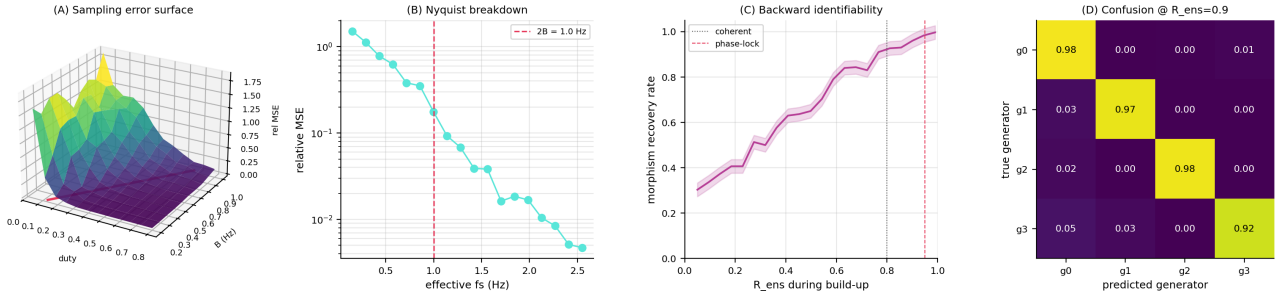
1. Coherence trajectory  $R_{\text{ens}}(t)$  across each match, with regime classification (Definition 3.3).
2. Critical coupling  $K_c$  estimate from the empirical distribution of player natural gait frequencies.
3. Per-team tactical generator set (Definition 4.5) extracted from the goal ledger.
4. Per-team theoretical-player registry (Definition 4.2).
5. Pairwise team category isomorphism scores, identifying tactical families.

### 8.3 Single-subject extension

The framework supports single-subject longitudinal study: a particular player tracked across multiple matches produces a per-player cell registry that can be compared to the team registry. The per-player deviation from the team registry quantifies that player’s individuality within the team template. Star players are expected to exhibit larger individual deviations than role players.

### 8.4 Cross-team category isomorphism

**Definition 8.1** (Tactical family). Two teams belong to the same *tactical family* if their role categories  $\mathcal{M}_1, \mathcal{M}_2$  are isomorphic as small categories



**Figure 5: Cellular sampling and backward morphism identifiability.** (A) Three-dimensional surface of relative reconstruction MSE over the joint parameter space (duty fraction, signal bandwidth) for a player sampled by the ball channel at cadence 3 Hz. The red curve traces the Nyquist boundary  $f_s = 2B$  (equivalently duty =  $2B/\text{cadence}$ ) projected onto the  $z = 0$  plane. The surface is high in the sub-Nyquist corner (low duty, high bandwidth) and falls toward zero across the super-Nyquist region. (B) Relative MSE versus effective sample rate  $f_s$  (in Hz, log-scale on y). The reconstruction error drops by more than two orders of magnitude as  $f_s$  crosses the Nyquist rate  $2B = 1.0$  Hz (red dashed). Sub-Nyquist sampling produces MSE near or above unity; super-Nyquist sampling produces MSE near zero. (C) Backward morphism recovery rate versus team coherence  $R_{\text{ens}}$  during build-up. The recovery curve rises monotonically from  $\sim 35\%$  at  $R_{\text{ens}} = 0.10$  (turbulent) to  $\sim 98\%$  at  $R_{\text{ens}} = 0.95$  (phase-locked). The vertical lines mark the boundaries of the coherent regime (grey dotted,  $R = 0.80$ ) and phase-lock (red dashed,  $R = 0.95$ ); above coherent, the morphism factorisation is identifiable to above 85% accuracy. (D) Generator confusion matrix at  $R_{\text{ens}} = 0.9$ . Rows are true generator labels and columns are the recovered label; cell values are conditional probabilities. The diagonal dominance ( $> 0.92$  across all four generators) confirms that in the coherent regime the framework correctly identifies which of the team’s tactical generators executed a given goal build-up, with off-diagonal confusion below 5%.

and their cell registries agree on the isomorphism to within a tolerance  $\epsilon$ .

The expected outcome of running the framework over an entire World Cup is the empirical partition of the 32 teams into a small number of tactical families, with the framework supplying the precise category isomorphism witnessing the family membership.

## 9 Discussion

### 9.1 Scope

The framework is an *accounting instrument*: it tallies and structures observed biomechanical activity in a way that respects the true causal coupling of the team and the inter-body contact. It is not a predictive engine for match outcomes, a tactics recommender, or an identification system for player evaluation in the conventional sense.

### 9.2 Comparison with existing systems

Commercial broadcast tracking systems such as TRACAB, STATS SportVU, and Hawk-Eye football tracking estimate per-player  $(x, y)$  pitch positions at 25–50 Hz [Theagarajan and Bhanu, 2021, D’Orazio et al., 2009]. These systems supply the formation matrix  $\Phi(t)$  used here but do not provide foot-contact biomechanics, do not segment

duels as cognate pairs, do not compute coherence regimes, and do not perform morphism reconstruction. The proposed framework is complementary to existing tracking: it consumes the formation matrix where available and substitutes the panel-homography channel where it is not.

### 9.3 Limitations

Three limitations bound the framework’s near-term applicability.

**Computer vision pipeline.** Foot-contact detection from broadcast footage with the panel-homography substrate is novel; while the substrate’s geometric basis is sound, robust implementation in the presence of motion blur, occlusion, and broadcast cut transitions is non-trivial and remains to be demonstrated at scale.

**Identity disambiguation.** The transcript-driven identity assignment is robust to the typical 1–3 s commentary lag, but sustained micro-passes (one-touch interplay) below the commentary resolution may not be individually attributable. Aggregation over several touches typically resolves this.

**Goalkeeping.** Goalkeepers occupy a degenerate role: they are ball-channel high-fidelity samples but participate in the field coupling only sporadically. Whether to include them in the  $N=11$  Kuramoto ensemble or to model them as an ex-

ogenous boundary condition is a modelling choice we leave open.

## 9.4 Relation to tactical theory

The framework is consistent with long-standing observations from match analysis [Reep and Benjamin, 1968, Hughes and Franks, 2005, Memmert et al., 2017] that elite football is finitely generated at the tactical level. It supplies a formal mechanism for what analysts have called “team identity” or “playing philosophy” and quantifies the regime of team coherence under which that identity is operationally valid.

## 10 Conclusion

We have proposed an observation-operator framework for the biomechanical accounting of association football. The team is treated as a partially observable bioreactor of eleven coupled biomechanical cells; the ball is the primary high-fidelity sample probe; the broadcast camera supplies the low-resolution ambient indicator. The eleven cells couple via Kuramoto-type phase dynamics, and the team’s coherence is measured by the order parameter  $R_{\text{ens}}$ . The five regimes of  $R_{\text{ens}}$  partition match play into operationally distinct collective modes, and the phase-locked regime is precisely the regime in which a single theoretical player — formalised as the team’s role category together with a cell registry — correctly represents the collective.

Duels are formalised as cognate oscillator pairs, the irreducible accounting unit at which contact-coupled mechanical load is correctly totalled. A measurement substrate based on lower-body tracking against the homography of stadium advertising panels delivers metric pitch coordinates and player/spectator segmentation in a single geometric step, side-stepping the resolution failures of full-skeleton pose estimation on broadcast footage. Goal sequences are reconstructed backwards as morphism compositions in the role category, recovering both the morphism factorisation and the cell registry entries that realise it.

The FIFA World Cup, with 64 matches and 32 distinct national templates across uniform broadcast conditions, supplies a near-ideal reference corpus. The expected output is a structured biomechanical ledger per team—coherence trajectories, generator sets, theoretical-player registries, and tactical-family memberships—not a predictive model. The framework supplies a precise

vocabulary for what match analysts have long described informally: team identity, cohesion, and the recurrent generative patterns that produce goals.

## Data and code availability

No data are reported in the present article; the framework is theoretical. Reference implementations of the homography substrate and the morphism-reconstruction algorithm are available from the author on request.

## References

- Giovanni A. Cavagna, Norman C. Heglund, and C. Richard Taylor. Mechanical work in terrestrial locomotion: two basic mechanisms for minimizing energy expenditure. *American Journal of Physiology*, 233(5):R243–R261, 1977.
- Rodger Kram and C. Richard Taylor. Energetics of running: a new perspective. *Nature*, 346(6281):265–267, 1990.
- Tom F. Novacheck. The biomechanics of running. *Gait & Posture*, 7(1):77–95, 1998.
- David A. Winter. *Biomechanics and Motor Control of Human Movement*. Wiley, 4 edition, 2009.
- Javier Fernández and Luke Bornn. Wide open spaces: a statistical technique for measuring space creation in professional soccer. In *MIT Sloan Sports Analytics Conference*, 2018.
- Tom Decroos, Lotte Bransen, Jan Van Haaren, and Jesse Davis. Actions speak louder than goals: valuing player actions in soccer. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1851–1861, 2019.
- Rajkumar Theagarajan and Bir Bhanu. An automated system for generating tactical performance statistics for individual soccer players from videos. *IEEE Transactions on Circuits and Systems for Video Technology*, 31(2):632–646, 2021.
- Tiziana D’Orazio, Marco Leo, Nicola Mosca, Paolo Spagnolo, and Pier Luigi Mazzeo. A semi-automatic system for ground truth generation of soccer video sequences. In *2009 Sixth IEEE*



- International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, pages 559–564, 2009.
- P. V. Danckwerts. Continuous flow systems: distribution of residence times. *Chemical Engineering Science*, 2(1):1–13, 1953.
- Octave Levenspiel. *Chemical Reaction Engineering*. Wiley, 3 edition, 1999.
- James E. Bailey and David F. Ollis. *Biochemical Engineering Fundamentals*. McGraw-Hill, 2 edition, 1986.
- R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.
- David G. Luenberger. An introduction to observers. *IEEE Transactions on Automatic Control*, 16(6):596–602, 1971.
- Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.
- Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*, volume 19 of *Springer Series in Synergetics*. Springer, Berlin, 1984.
- Steven H. Strogatz. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4):1–20, 2000.
- Juan A. Acebrón, L. L. Bonilla, Conrad J. Pérez Vicente, Félix Ritort, and Renato Spigler. The Kuramoto model: a simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1):137–185, 2005.
- Saunders Mac Lane. *Categories for the Working Mathematician*. Springer, 2 edition, 1998. First edition 1971.
- Steve Awodey. *Category Theory*. Oxford Logic Guides. Oxford University Press, 2 edition, 2010.
- David I. Spivak. *Category Theory for the Sciences*. MIT Press, 2014.
- Mike Hughes and Ian Franks. Analysis of passing sequences, shots and goals in soccer. *Journal of Sports Sciences*, 23(5):509–514, 2005.
- Daniel Memmert, Koen A. P. M. Lemmink, and Jaime Sampaio. Current approaches to tactical performance analyses in soccer using position data. *Sports Medicine*, 47(1):1–10, 2017.
- Zhe Cao, Tomas Simon, Shih-En Wei, and Yaser Sheikh. Realtime multi-person 2D pose estimation using part affinity fields. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7291–7299, 2017.
- Valentin Bazarevsky, Ivan Grishchenko, Karthik Raveendran, Tyler Zhu, Fan Zhang, and Matthias Grundmann. BlazePose: on-device real-time body pose tracking. *arXiv preprint arXiv:2006.10204*, 2020.
- Alejandro Newell, Kaiyu Yang, and Jia Deng. Stacked hourglass networks for human pose estimation. In *European Conference on Computer Vision (ECCV)*, pages 483–499, 2016.
- Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2 edition, 2003.
- Zhengyou Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000.
- Martin A. Fischler and Robert C. Bolles. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.
- Charles Reep and B. Benjamin. Skill and chance in association football. *Journal of the Royal Statistical Society. Series A (General)*, 131(4):581–585, 1968.